

Section-c

Q:3(i)

A:3(i)

$$a(x^2 + 3x - 4)(x^2 + 3x + 5) + 8$$

let $x^2 + 3x = a$

$$\begin{aligned} &= (a - 4)(a + 5) + 8 \\ &= a(a + 5) - 4(a + 5) + 8 \\ &= a^2 + 5a - 4a - 20 + 8 \\ &= a^2 + a - 12 \\ &= a^2 + 4a - 3a - 12 \\ &= a(a + 4) - 3(a + 4) \\ &= (a - 3)(a + 4) \end{aligned}$$

Putting $x^2 + 3x$ back.

$$(x^2 + 3x - 3)(x^2 + 3x + 4) \text{ ans}$$

Q:3(ii)

A:3(ii)

HCF

$$P = 2m^2 - 2m, Q = 4m^3 - 4m^2n^2, R = 2m^3 - 4m^2n^2$$

$$\textcircled{1} P = 2m^2 - 2mn$$

$$= 2m(m-n)$$

$$\textcircled{2} Q = 4m^4 - 4m^2n^2$$

$$= 4m^2(m^2 - n^2)$$

$$\textcircled{3} R = 2m^3 - 4m^2n + 2mn^2$$

$$= 2m(m^2 - 2mn + n^2)$$

$$= 2m(m-n)^2$$

H.C.F =

$$P = 2m(m-n)$$

$$Q = 2^2 \times m^2(m-n)(m+n)$$

$$R = 2m(m-n)^2$$

$$\text{H.C.F} = 2m(m-n) \text{ Ans.}$$

Q.3(iii)

A:3(iii)

$$\text{L.C.M} = 2x^2 - 12xy + 16y^2, (x^2 - 6xy + 8y^2), 3x^2 - 11y^2$$

$$2x^2 - 12xy + 16y^2 = 2(x^2 - 6xy + 8y^2)$$

$$= 2(x^2 - 4xy - 2xy + 8y^2)$$

$$= 2(x(x-4y) - 2y(x-4y))$$

$$= 2(x-2y)(x-4y)$$

$$\textcircled{2} x^2 - 6xy + 8y^2$$

$$x^2 - 4xy - 2xy + 8y^2$$

$$x(x - 4y) - 2y(x - 4y)$$

$$(x - 2y)(x - 4y)$$

$$3x^2 - 12y^2$$

$$= 3(x^2 - 4y^2)$$

$$= 3(x^2 - (2y)^2)$$

$$= 3(x - 2y)(x + 2y)$$

$$\therefore \text{L.C.M} = 2 \times 3 (x - 4y)(x - 2y)(x + 2y)$$

$$\text{H.C.F} =$$

$$P = 2(x - 4y)(x - 2y)$$

$$Q = (x - 2y)(x - 4y)$$

$$R = 3(x - 2y)(x + 2y)$$

$$\therefore \text{H.C.F} = (x - 4y)(x - 2y)$$

$$\textcircled{3} \text{ L.C.M} \times \text{H.C.F} = P \times Q$$

$$2 \times 3 (x - 4y)(x - 2y)(x + 2y) \times (x - 4y)(x - 2y)$$

$$= 6(x - 4y)^2(x - 2y)^2(x + 2y)$$

$$= P \times Q$$

$$2(x - 2y)(x - 4y) \times (x - 2y)(x - 4y)$$

$$= 2(x-2y)^2(x-4y)^2$$

$$L.C.M(p, q) \times H.C.F.(p, q) = p \times q$$

$$2(x-4y)(x-2y) \times (x-4y)(x-2y) = \frac{2(x-4y)^2(x-2y)^2}{(x-4y)(x-2y)}$$

$$= 2(x-4y)^2(x-2y)^2 = 2(x-4y)^2(x-2y)^2$$

— Q: 3(iv) —

— A: 3(iv) —

$$\frac{-2x}{(x+3)} + \frac{3}{(3-x)} - \frac{(8x-12)}{(x^2-9)}$$

$$\frac{-2x}{x+3} + \frac{3}{-(3-x)} - \frac{(8x-12)}{(x^2-3^2)}$$

$$= \frac{-2x}{x+3} + \frac{3}{-(x-3)} - \frac{8x-12}{(x-3)(x+3)}$$

$$= \frac{-2x}{x+3} - \frac{3}{x-3} - \frac{8x-12}{(x-3)(x+3)}$$

$$= \frac{-2x(x+3)}{(x+3)(x-3)}$$

$$= \frac{-2x(x+3)}{x+3} - 3 \frac{(x+3)}{x-3} - \frac{(8x-12)}{(x-3)(x+3)}$$

$$= \frac{-2x^2 + 6x - 3x^2 - 9x - 8x + 12}{(x+3)(x-3)}$$

$$= \frac{-2x^2 - 5x + 12}{(x+3)(x-3)}$$

$$= \frac{-2x^2 - 6x + x + 3}{(x+3)(x-3)}$$

$$= \frac{-2x(x+3) + 1(x+3)}{(x+3)(x-3)}$$

$$= \frac{(-2x+1)(x+3)}{(x+3)(x-3)}$$

$$= \frac{(2x+1)(x-3)}{(x+3)(x-3)}$$

$$= \frac{(2x+1)}{(x-3)}$$

Section :- B

Q:2(i) —

A:2(i) —

$$2xy^3 = 6xy^2 + 2xy^3$$

$$= 2xy^2(xy - 3x + y)$$

Q:2(ii)(a) —

A:2(ii)(a) —

$$1 - x^2 - 2xy - y^2$$

$$= -1(x^2 - 2xy - y^2)$$

$$= -1(x^2 - y^2)$$

Q:2(iii) —

A:2(iii) —

$$x^4 + 4m^4$$

$$= (x^2)^2 + (2m^2)^2$$

$$= (x^2)^2 + \frac{(2m^2)^2}{(2m^2)}(x^2)(2m^2) + (2m^2)^2$$

$$= x^4 + 2x^2m^2 + 2m^4$$

$$= (x^2 + 2m^2)^2$$

Q:2(iv) —

A:2(iv) —

$$y^2 - 7y + 12$$

$$= y^2 - 3y - 4y + 12$$

$$= y(y - 3) - 4(y - 3)$$

$$= (y - 4)(y - 3)$$

Q:2(v)

A:2(v)

$$(x)^3 + (3y)^3 \\ (x+3y)(x^2 - 2(x)(3y) + (3y)^2) \\ (x+3y)(x^2 - 6xy + 9y)$$

Q:2(vi)

A:2(vi)

$$4n^2(2n-15) - 25(5-6n) \\ 8n^3 - 60n^2 - 125 - 150n \\ 2n^2(4n^2 - 30) - 5(25 - 30n)$$

Q:2(vii)

H.C.F.:-

$$\textcircled{1} (x+y)^2 \\ = (x)^2 + (x)(y) + (y)^2 \\ = \cancel{x+y} (x+y)^2 \text{ or } (x+y)(x+y)$$

$$\textcircled{2} x^2 - y^2 \\ = (x)^2 - (y)^2 \\ (x+y)(x-y)$$

$$\text{H.C.F.} = xy$$

Q:2(viii)(or)

A:2(vii)(or)

(P.T.O)

H.C.F.:

① $x^2 - 49$

$$(x^2 - 7^2)$$

$$(x+7)(x-7)$$

② $x^2 - 4x - 21$

$$x^2 - 7x + 3x - 21$$

$$x(x-7) + 3(x-7)$$

$$(x+3)(x-7)$$

H.C.F. = $x-7$.

Q: 2 (iv) (or)
A: 2 (iv) (or)

① $x^2 + x - 6$

$$= x \times x + x + 2 \times 3$$

② $x^2 + x - 2$

$$x \times x + x + -2$$

③ $x^2 - 4x + 3$

$$x \times x - 2x + 2 + 3$$

L.C.M. = $x^2 - x + 2x + 2 - 2 + 3$

— Q.2 (n) (or) —

— A : 2 (n) (or) —

$$\frac{(r^2 - s^2)}{r} \div \frac{(r - s)}{s}$$

$$\frac{r^2 - s^2}{r} \times \frac{s}{r - s}$$

$$\frac{(r + s)(r - s)}{r} \times \frac{s}{r - s}$$

$$\frac{r + s}{r}$$

— Q.2 (ni) —

— A : 2 (ni) —

$$4a^2 + 12ab + 9b^2$$
$$(2a)^2 + (2a)(3b) + (3b)^2$$
$$= (2a + 3b)^2$$

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Mathematics — SSC-I (9th Grade)

Syllabus: Chapter 4 — Factorization and Algebraic Manipulation

Total Marks: 65

Time Allowed: 2h 30m

Version: 1

General Instructions:

- Write your name, roll number, and section clearly on the answer sheet.
- Section A (MCQs):** Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
- Section B (Short Questions):** Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
- Section C (Long Questions):** Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
- Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The factorization of $ka + kb + kc$ is:	$k(a - b + c)$	$k(a + b + c)$	$k(a + b - c)$	$(k + a)(b + c)$	☐ ☐ ☒ ☐
2	The factorized form of $a^2 - b^2$ is:	$(a + b)^2$	$(a - b)^2$	$(a + b)(a - b)$	$(a - b)(a - b)$	☐ ☐ ☒ ☐
3	The factorized form of $a^4 + a^2b^2 + b^4$ is:	$(a^2 + b^2)^2$	$(a^2 + ab + b^2)(a^2 - ab + b^2)$	$(a^2 - b^2)^2$	$(a + b)^4$	☐ ☒ ☐ ☐
4	The factors of $x^2 - 9x + 8$ are:	$(x - 1)(x - 8)$	$(x + 1)(x - 8)$	$(x - 1)(x + 8)$	$(x - 2)(x - 4)$	☒ ☐ ☐ ☐
5	$a^3 + b^3$ is equal to:	$(a + b)(a^2 + ab + b^2)$	$(a - b)(a^2 + ab + b^2)$	$(a + b)(a^2 - ab + b^2)$	$(a + b)^3$	☐ ☒ ☐ ☐
6	The expression $a^3 + 3a^2b + 3ab^2 + b^3$ is equal to:	$(a - b)^3$	$(a + b)^3$	$(a + b)(a^2 - ab + b^2)$	$(a + b)^2$	☐ ☒ ☐ ☐
7	HCF uses the <i>lowest</i> power of each common factor, while LCM uses the:	Lowest power of every factor	Highest power of common factors only	Highest power of every factor	Product of all uncommon factors	☐ ☒ ☐ ☐
8	If HCF of two polynomials P and Q is H, then LCM equals:	$P \times Q \times H$	$P \times Q + H$	$H + (P \times Q)$	$P + Q - H$	☐ ☒ ☐ ☐
9	An algebraic fraction is an expression of the form P/Q where P and Q are polynomials and:	$Q = 0$	$P = 0$	$Q \neq 0$	$P = Q$	☐ ☐ ☒ ☐
10	The square root of $4a^2 + 12ab + 9b^2$ is:	$\pm(2a + 3b)$	$\pm(2a - 3b)$	$\pm(4a + 9b)$	$\pm(a + b)$	☒ ☐ ☐ ☐
11	To add algebraic fractions with different denominators, we first find the:	HCF of denominators	LCM of denominators	HCF of numerators	LCM of numerators	☐ ☒ ☐ ☐
12	Factorization is essentially the <i>opposite</i> of which process?	Division	Subtraction	Multiplication	Integration	☐ ☐ ☒ ☐

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question OR its alternative.